

Lecture 26E

COVID-19 Virus Content

Dr. Neil Voss
April 28, 2022

Definitions

- Outbreak – a sudden rise in the incidence of a disease
- Pandemic – occurring over a wide area and affecting an exceptionally high proportion of the population
 - a pandemic outbreak of a disease – an outbreak of a disease that occurs over a wide geographic area and affects an exceptionally high proportion of the population
- Epidemic – affecting or tending to affect a disproportionately large number of individuals within a population, community, or region at the same time
 - an outbreak of epidemic disease – an outbreak of disease that spreads quickly and affects many individuals at the same time

Naming System

- the infectious virus
 - severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)
 - named by the Coronavirus Study Group (CSG) of the International Committee on Taxonomy of Viruses (ICTV)
- the disease and it's symptoms
 - coronavirus disease 2019 (COVID-19)
 - named by the World Health Organization (WHO)

COVID-19 is
a pandemic outbreak

an outbreak of a disease that occurs over a wide geographic area and affects an exceptionally high proportion of the population

Contagion (2011)



Contagion Movie compared to COVID-19

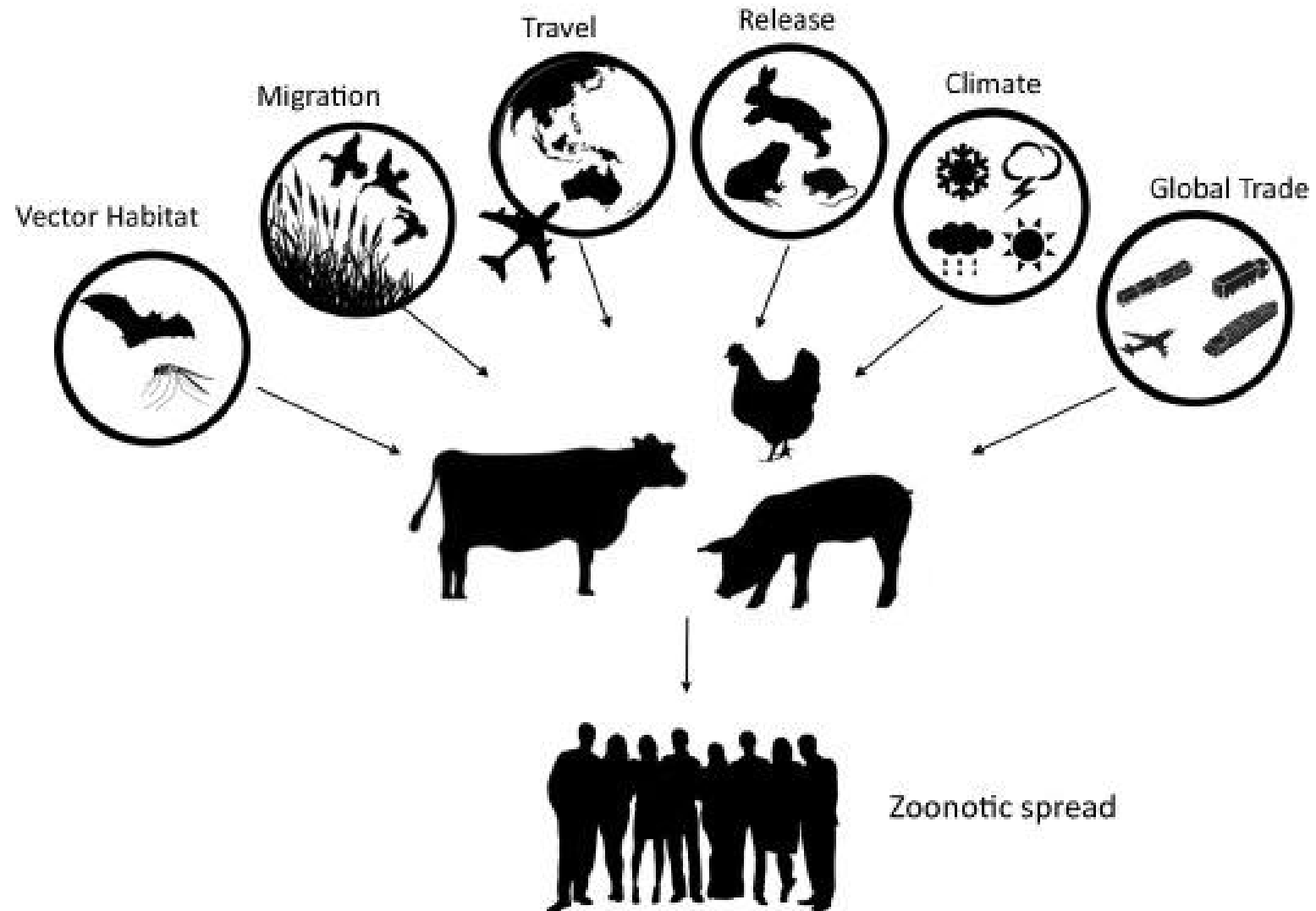
- the movie virus jumped from bat to pig to people — SARS-CoV-2 in all likelihood jumped from bat to human
- imaginary Contagion disease kills 25-30% of those infected
 - a magnitude beyond the estimated ~2% death rate in the current outbreak
 - The case fatality rate of the seasonal flu in the US is around 0.1% to 0.2%, while the case fatality rate for COVID-19, measured in the cited study, was 2.3%.

Zoonosis

Birth of an Outbreak

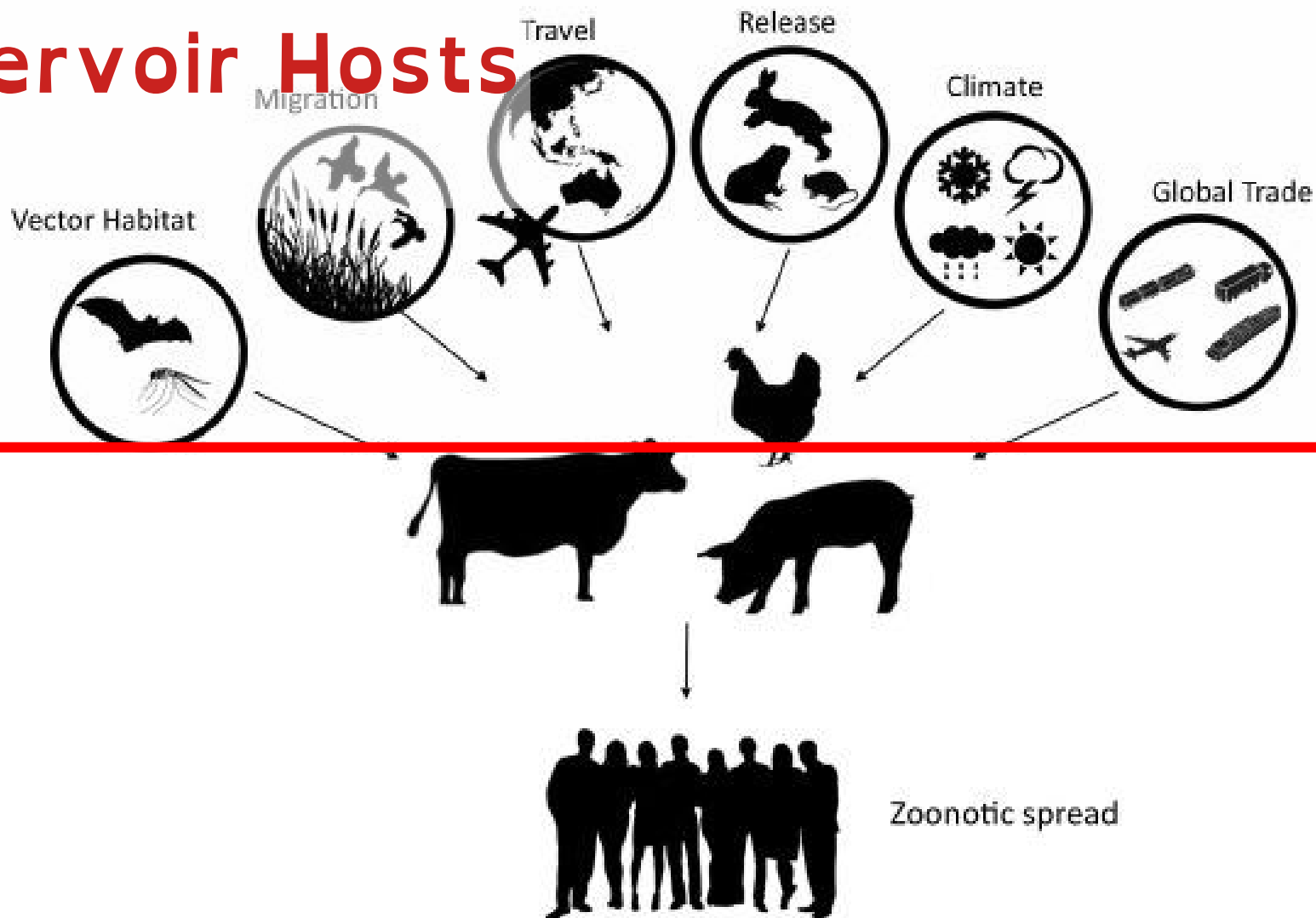
- "If you cut down trees, it changes the behavior of bats. The bats interact with swine, [which are] being raised for consumption, and humans then interact with infected swine as part of food preparations."
- Rebecca Katz, director of the Center for Global Health Science and Security at Georgetown University

Zoonosis

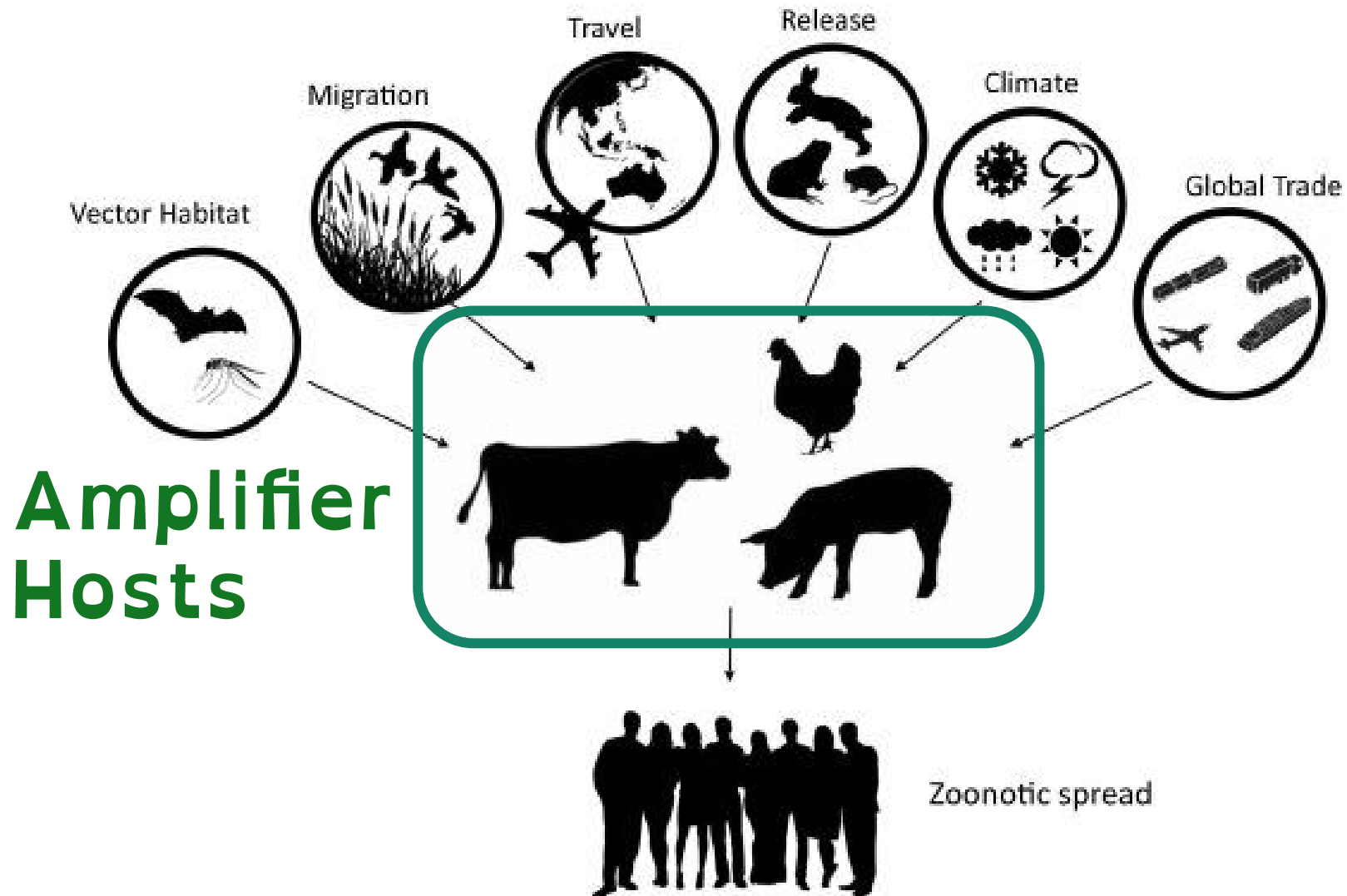


Zoonosis

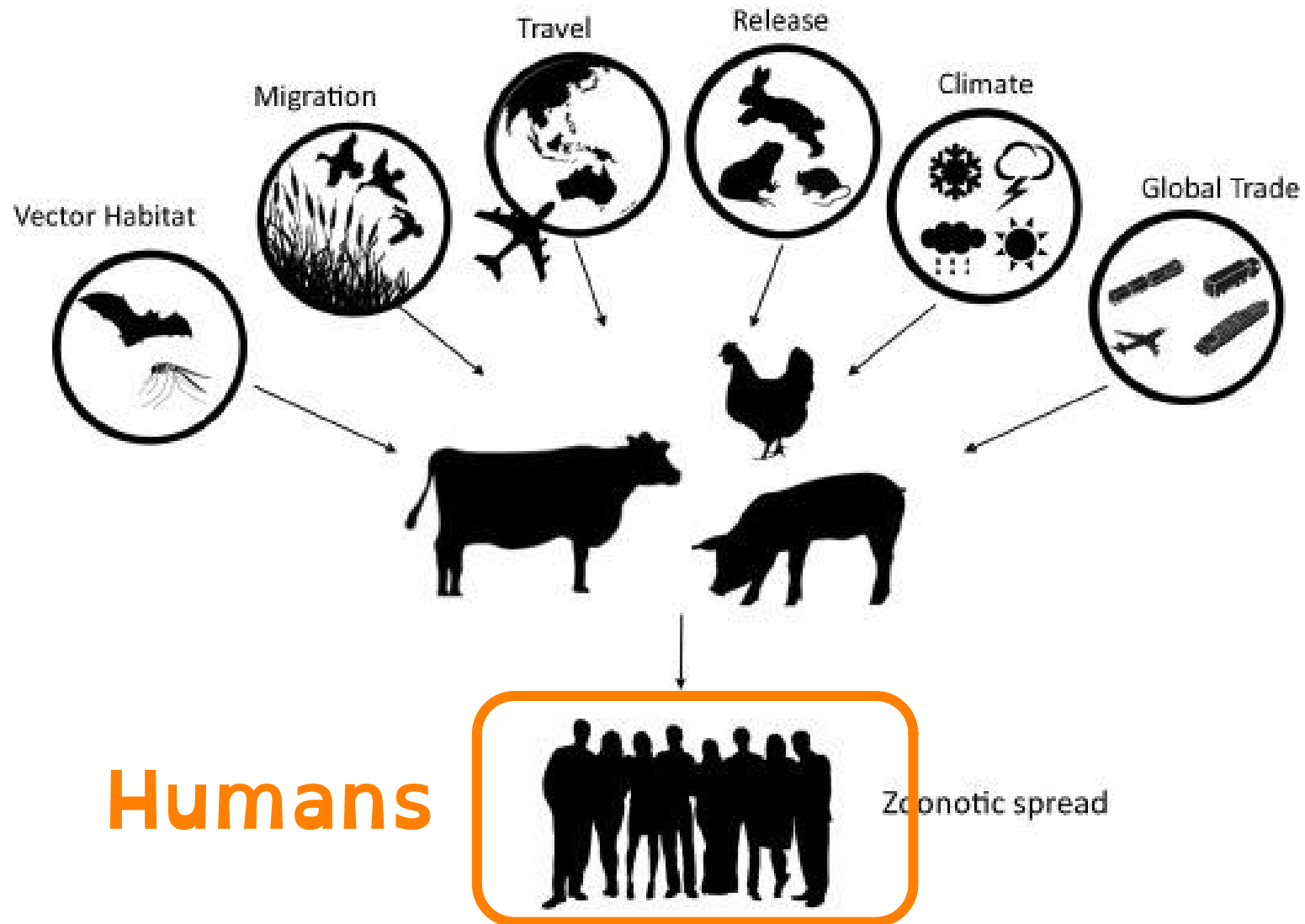
Reservoir Hosts



Zoonosis



Zoonosis



MEV1 vs SARS-CoV-2

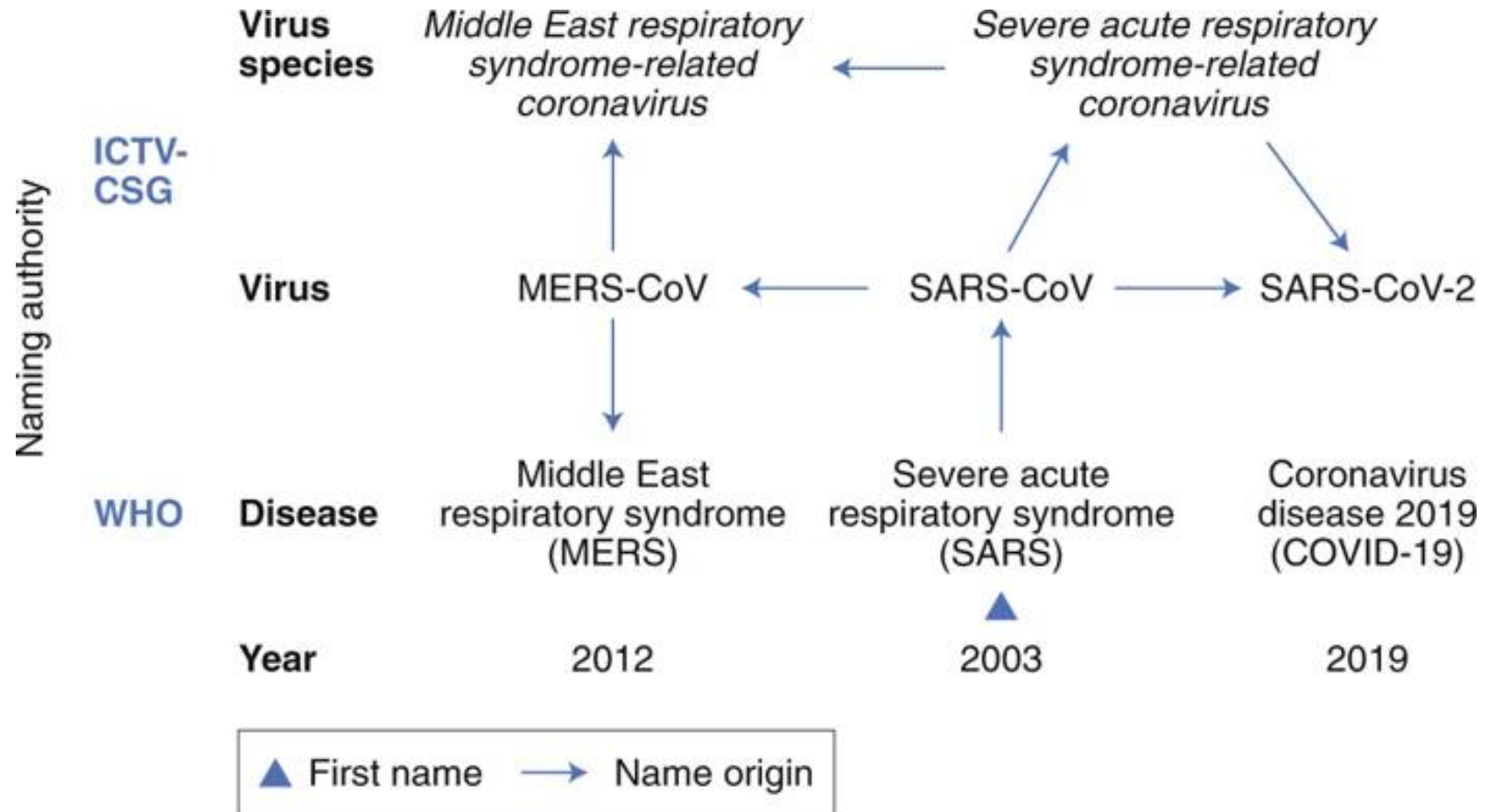
- Similar
 - originated in Asia
 - transferred from bats to humans
- Different
 - Final mortality rate of 25-30% in movie
 - Estimates range 1-4% for COVID-19

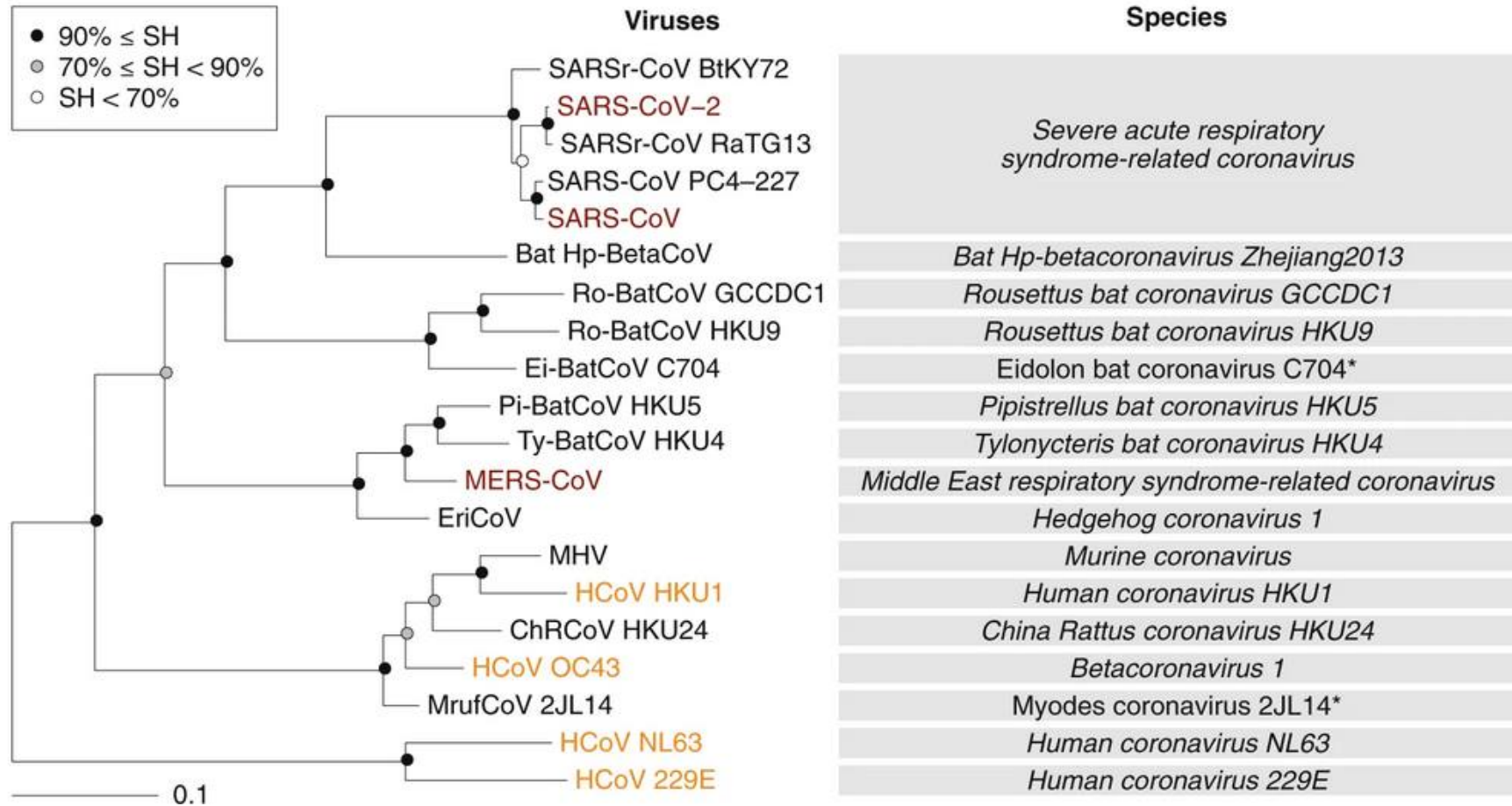
Fatality Rates

Disease	Estimated case fatality rate (CFR)
SARS-CoV1	10% Venkatesh and Memish (2004) Munster et al. (2020)
MERS-CoV	34% Munster et al. (2020)
Seasonal flu (US)	0.1% US CDC
Ebola	50% (40%; 2013-16 outbreak) WHO (2020) Shultz et al. (2016)
SARS-CoV2	1.82%
Contagion MEV-1	25-30%

SARS-CoV-2

- independent zoonotic transmission of SARS-CoV and SARS-CoV-2
- coronaviruses
 - a family of enveloped positive-strand RNA viruses infecting vertebrates
 - current classification of coronaviruses recognizes 39 species in 27 subgenera, five genera and two subfamilies
 - assigned using computational framework of comparative genomics
 - With respect to novelty, SARS-CoV-2 differs from the two other zoonotic coronaviruses, SARS-CoV and MERS-CoV,





<https://doi.org/10.1038/s41564-020-0695-z>

Movie Clip:
Contagion (2011)
Fomites

5. How is the virus transmitted? What are fomites? How many times a day do people touch their hands to their face?

- The virus is transmitted in the air or by touch.
 - The virus is transmitted when a healthy person touches something that a person with the virus touched.
 - The virus can also be transmitted in the air when they cough or sneeze directly into the air without covering their mouth with a sleeve or elbow.
- Fomites refer to the transmission of viruses to healthy individuals through surfaces like door knobs, water fountains, elevator buttons, and from other people who have the virus.
 - The average person touches their face 2000-3000 times a day.
- How does this compare to COVID-19?

Summary 1. What are common modes of transmission for viruses? List two.

- Being in contact with a person who is already infected
- touching surface or items that an infected person has already touched and touching your face with the hands

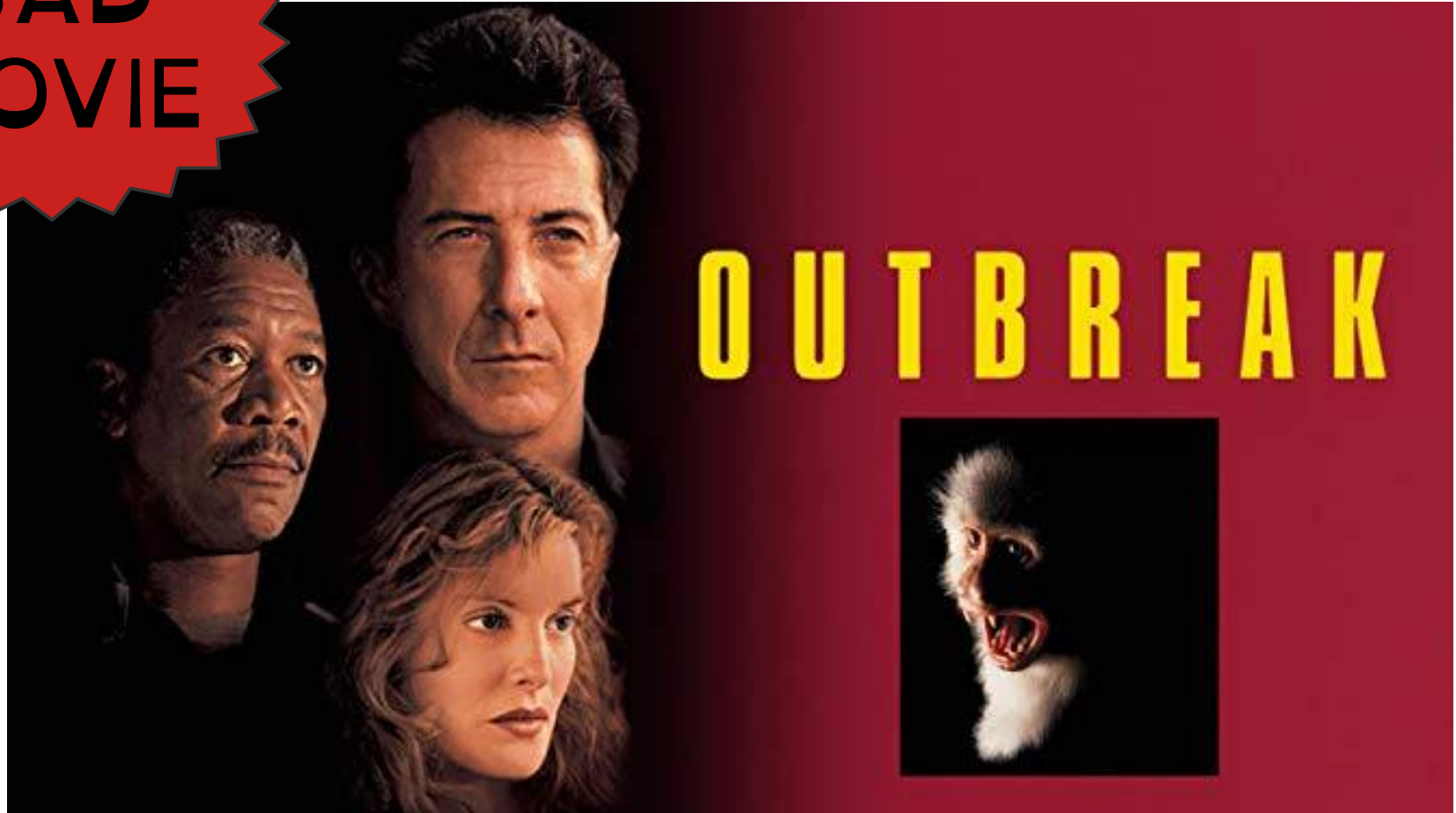
Contagion (2011)

**GOOD
MOVIE**



Outbreak (1995)

**BAD
MOVIE**



- Remember in **Outbreak** when Dustin Hoffman, in a full hazmat suit, went and met someone who had theoretically not been exposed?
- Yeah, that was dumb. He exposed that man to everything on his suit.
- “Truly, this guy should know better,” Amman says.

Movie Clip:
Outbreak (1995)
Hazmat suit public

R_0 factor

Movie Clip:
Contagion (2011)
R-naught

6. Explain what R_0 (R-naught) means?

- R_0 is referred to as the reproductive rate of the virus.
- It provides a way to track how many people get infected for every one person that gets sick.
 - It is essentially the way to study how a virus multiplies/replicates and how fast people get infected.

R_0 factor

- R_0 is pronounced “R naught” or “R zero” and often referred to as the basic reproduction number
- indicates how contagious an infectious disease is.
 - It’s also referred to as the reproduction number. As an infection spreads to new people, it reproduces itself.
- R_0 tells you the average number of people who will catch a disease from one contagious person.
 - If a disease has an R_0 of 18, a person who has the disease will transmit it to an average of 18 other people, as long as no one has been vaccinated against it or is already immune to it in their community.



More Contagious



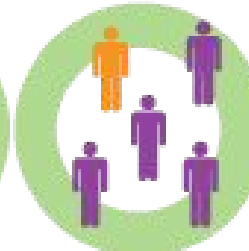
Hepatitis C (2)



Ebola (2)



HIV (4)



SARS (4)



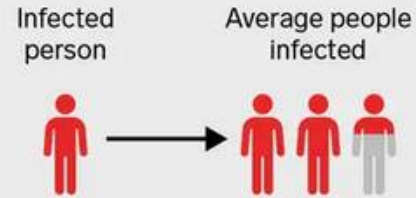
Mumps (10)



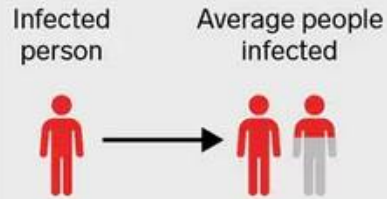
Measles (18)

The average number of people that one person with a virus infects, based on the R0 scale

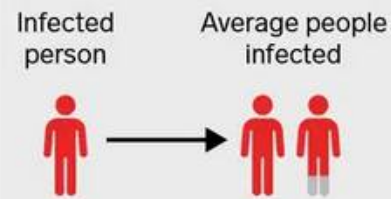
COVID-19: 2–2.5*



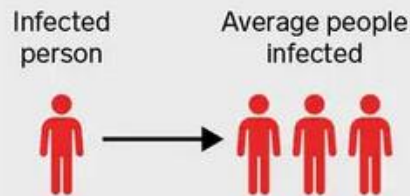
H1N1: 1.2–1.6



Ebola: 1.6–2



SARS: 2–4



MERS: 2.5–7.2**



*As of February 28, 2020 **R0 calculated solely during the 2015 outbreak in South Korea

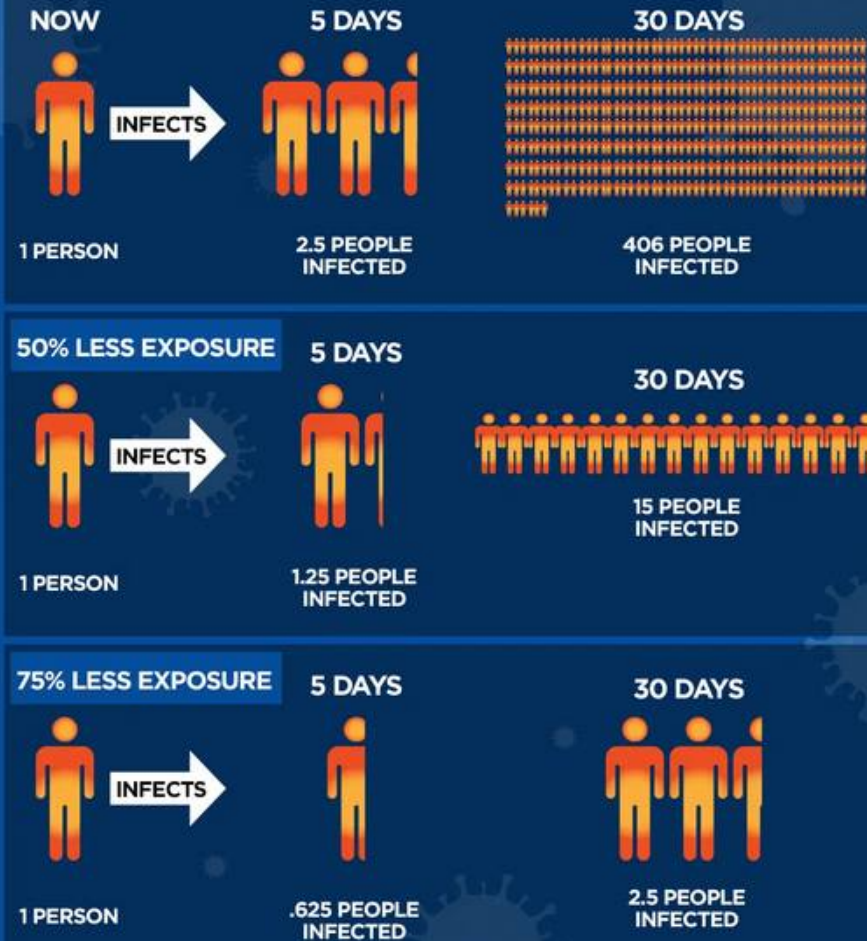
COVID-19 R_0

- COVID-19 has an estimated R_0 of 2-2.5, which is relatively high

Values of R_0 of well-known infectious diseases^[1]

Disease ⇅	Transmission ⇅	R_0 ⇅
Measles	Airborne	12–18
Diphtheria	Saliva	6–7
Smallpox	Airborne droplet	5–7
Polio	Fecal–oral route	5–7
Rubella	Airborne droplet	5–7
Mumps	Airborne droplet	4–7
Pertussis	Airborne droplet	5.5 ^[2]
HIV/AIDS	Sexual contact	2–5
SARS	Airborne droplet	2–5 ^[3]
COVID-19	Airborne droplet	1.4–3.9 ^{[4][5][6][7]}
Influenza (1918 pandemic strain)	Airborne droplet	2–3 ^[8]
Ebola (2014 Ebola outbreak)	Body fluids	1.5–2.5 ^[9]
MERS	Airborne droplet	0.3–0.8 ^[10]

The Math Behind Social Distancing



Credit:

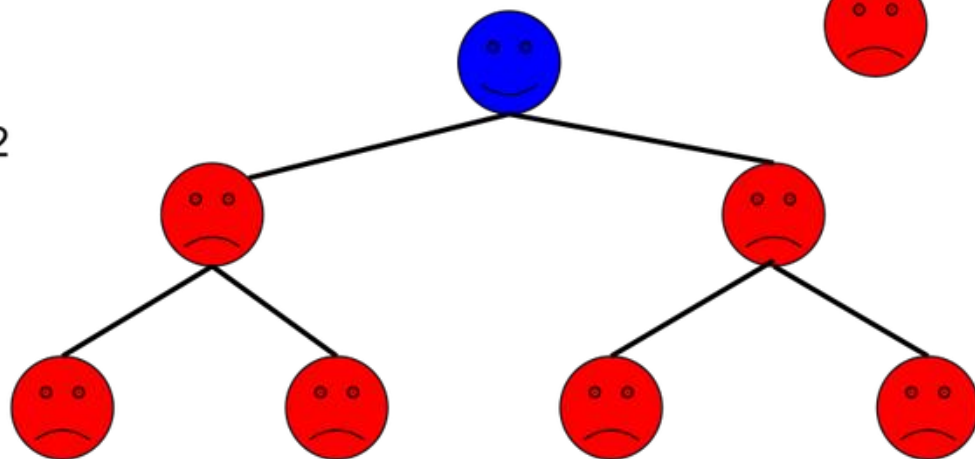
Robert A.J. Signer Ph.D.,

Assistant professor of Medicine at the University of California, San Diego

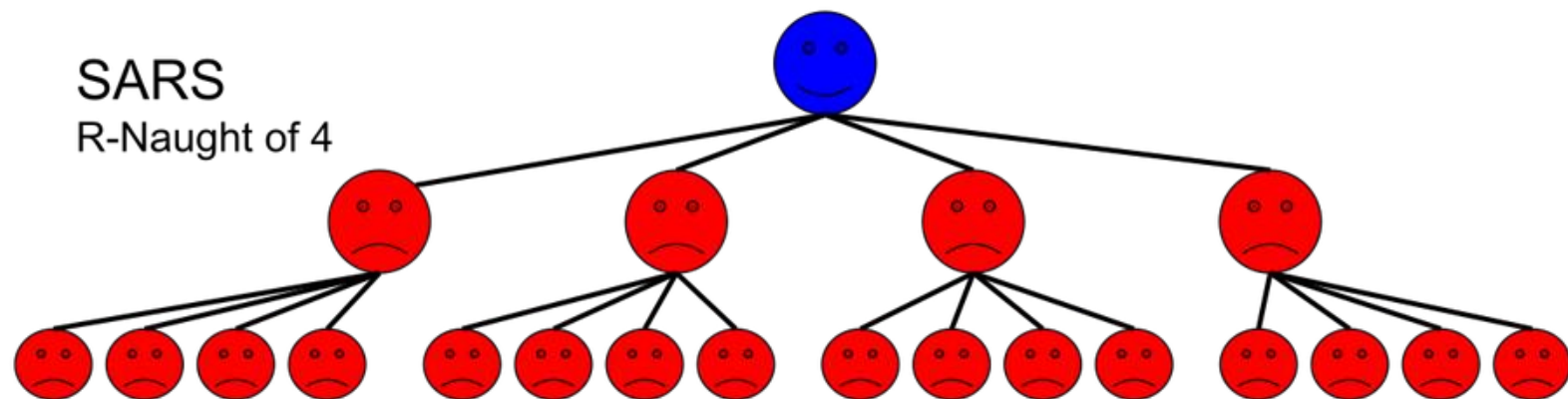
Gary Warsaw, Art Director



Ebola:
R-Naught of 2



SARS
R-Naught of 4



Incubation Period

Serial Interval and Incubation Period

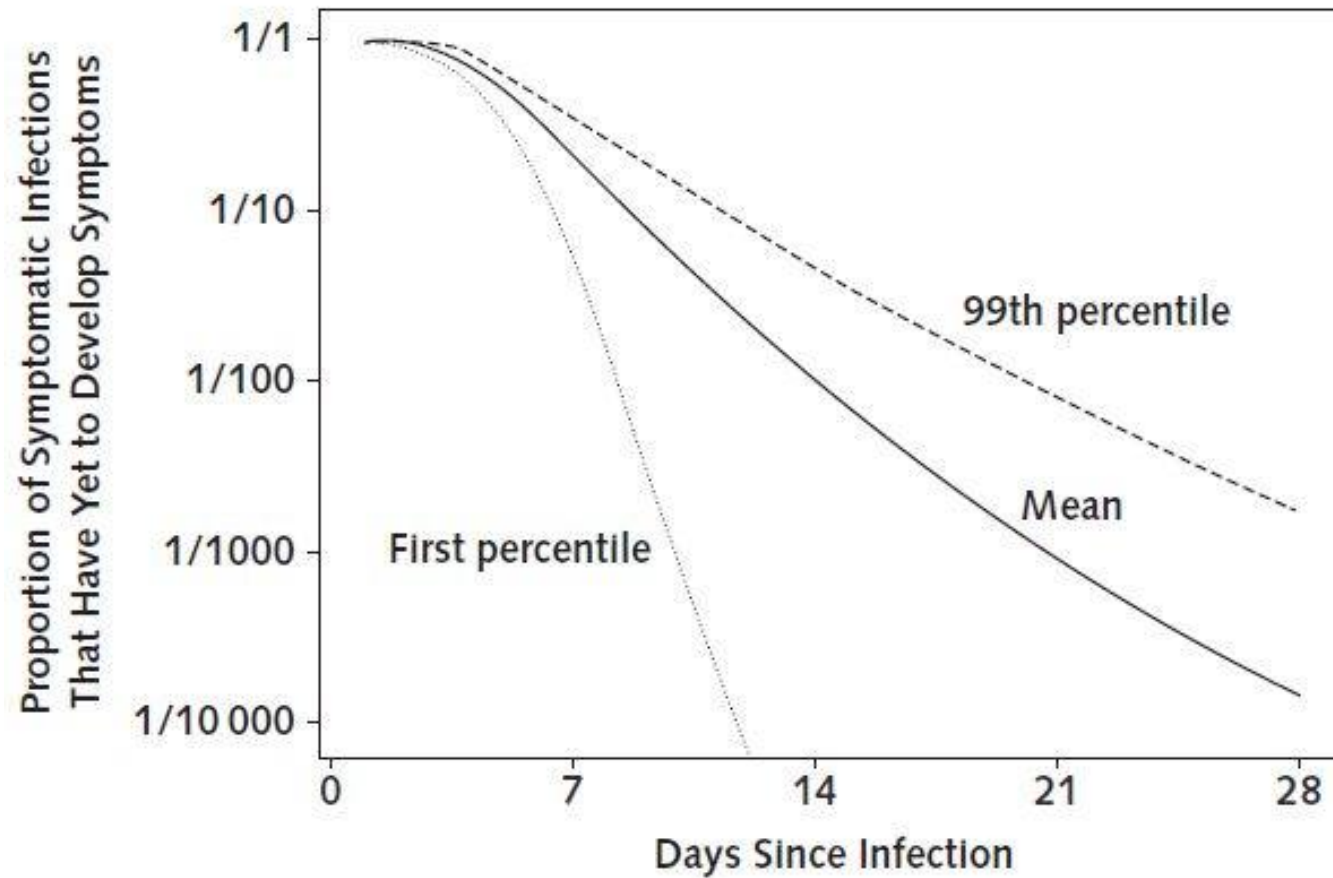
- **Serial interval**
 - average number of days between someone catching the virus and passing it on to someone else
 - estimated at 4.0 to 4.6 days
- **Incubation period**
 - time between exposure to disease/virus and showing first symptoms
 - estimated to be 5.1 days
- When the **serial interval** is shorter than the **incubation period**, pre-symptomatic transmission is more likely frequently than symptomatic transmission
 - In Singapore, transmission during the incubation period was estimated to account for 6.4 percent; in such cases, the exposures occurred one to three days prior to symptom development

Incubation Period

- Incubation period for the MER-1 virus is less than 10 days.
- The incubation period is the time from exposure to the causative agent until the first symptoms develop and is characteristic for each disease agent.
 - To determine the most likely period of exposure for an outbreak, you need to know the average incubation period for the disease and the range of incubation periods, which are the minimum and maximum reported incubation periods.

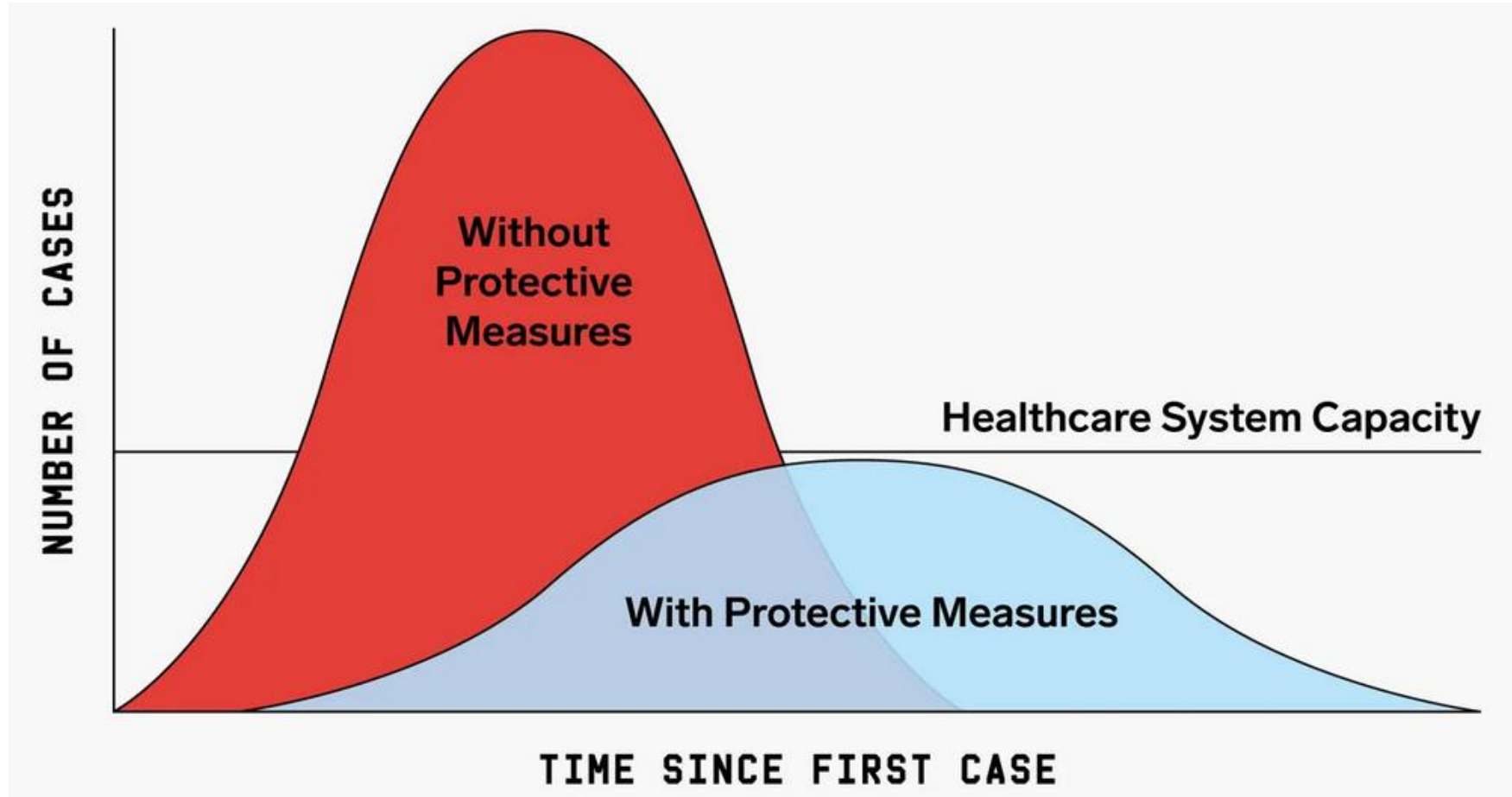
COVID-19 Incubation Rate

- The median incubation period was estimated to be 5.1 days
- 97.5% of those who develop symptoms will do so within 11.5 days
 - similar to SARS
 - There were 181 confirmed cases with identifiable exposure and symptom onset windows to estimate the incubation period of COVID-19
- These estimates imply that, under conservative assumptions, 101 out of every 10 000 cases (99th percentile, 482) will develop symptoms after 14 days of active monitoring or quarantine.



Flatten the Curve

Flatten the Curve

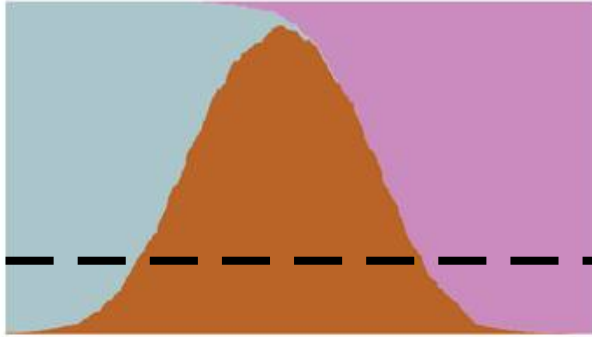


Corona Simulator

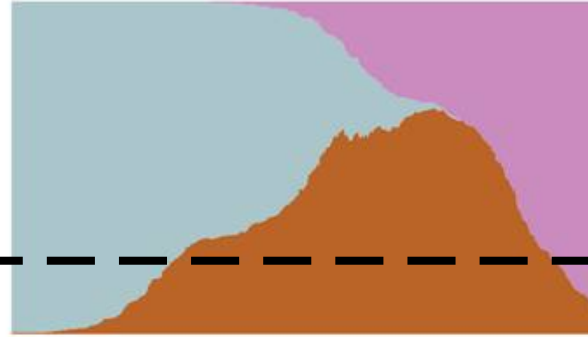
<https://www.washingtonpost.com/graphics/2020/world/corona-simulator/>

Comparison of the Results

Free-for-all

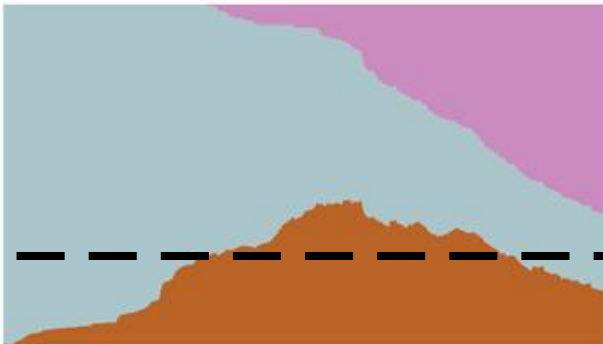


Attempted quarantine

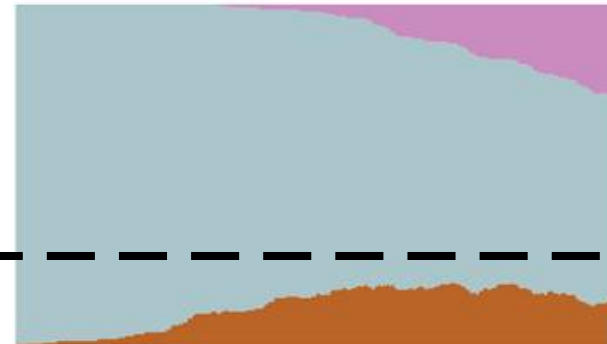


health care
capacity line

Moderate distancing

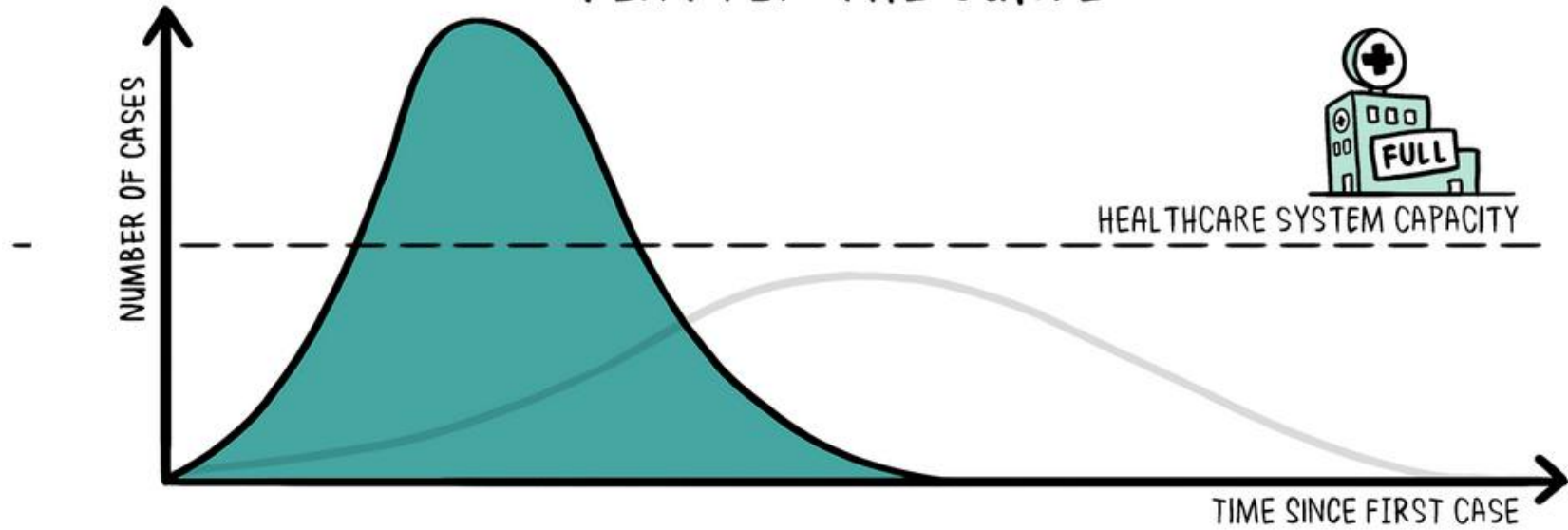


Extensive distancing

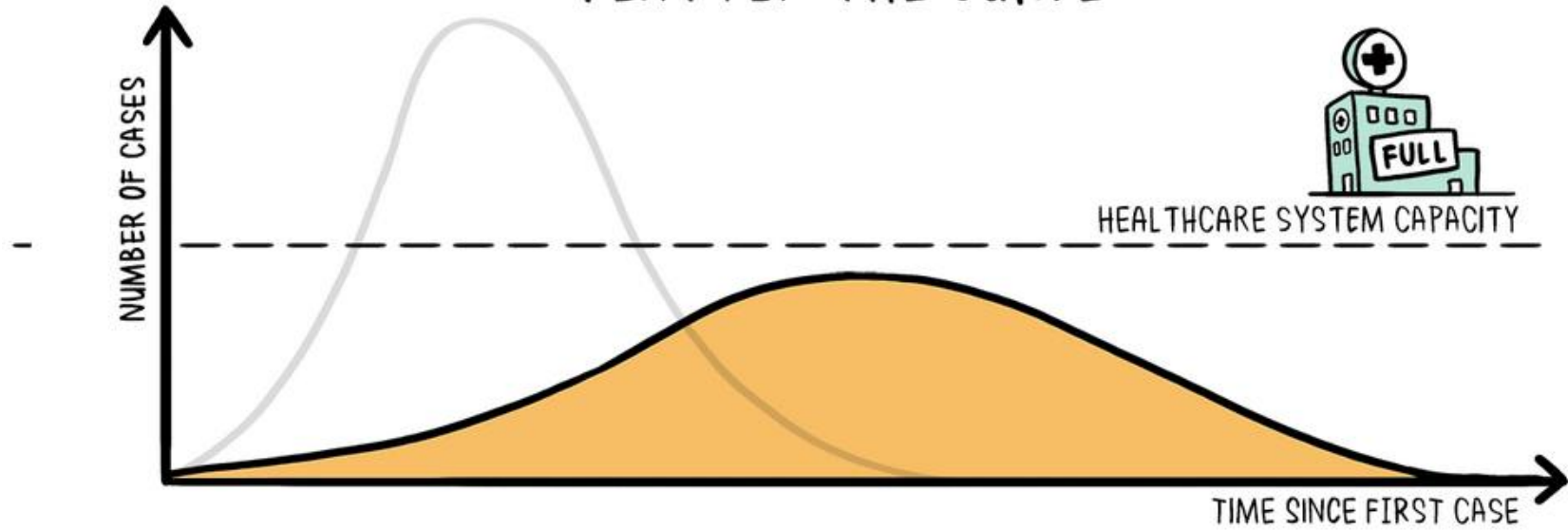


health care
capacity line

FLATTEN THE CURVE



FLATTEN THE CURVE



NOW



1 PERSON

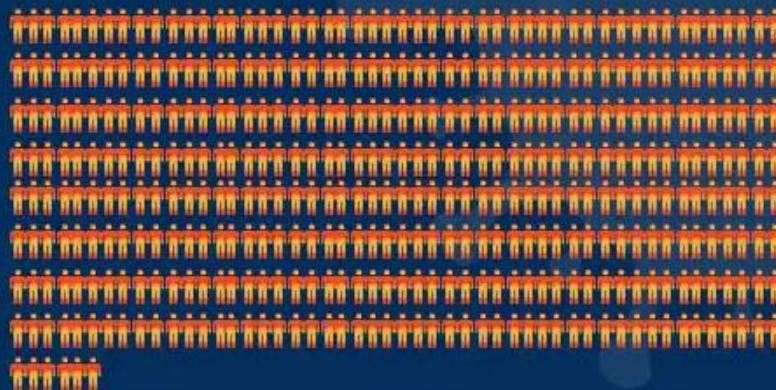
INFECTS

5 DAYS



**2.5 PEOPLE
INFECTED**

30 DAYS



**406 PEOPLE
INFECTED**

50% LESS EXPOSURE



1 PERSON

INFECTS



5 DAYS



**1.25 PEOPLE
INFECTED**

30 DAYS



**15 PEOPLE
INFECTED**

75% LESS EXPOSURE

5 DAYS

30 DAYS



INFECTS



1 PERSON

**.625 PEOPLE
INFECTED**

**2.5 PEOPLE
INFECTED**

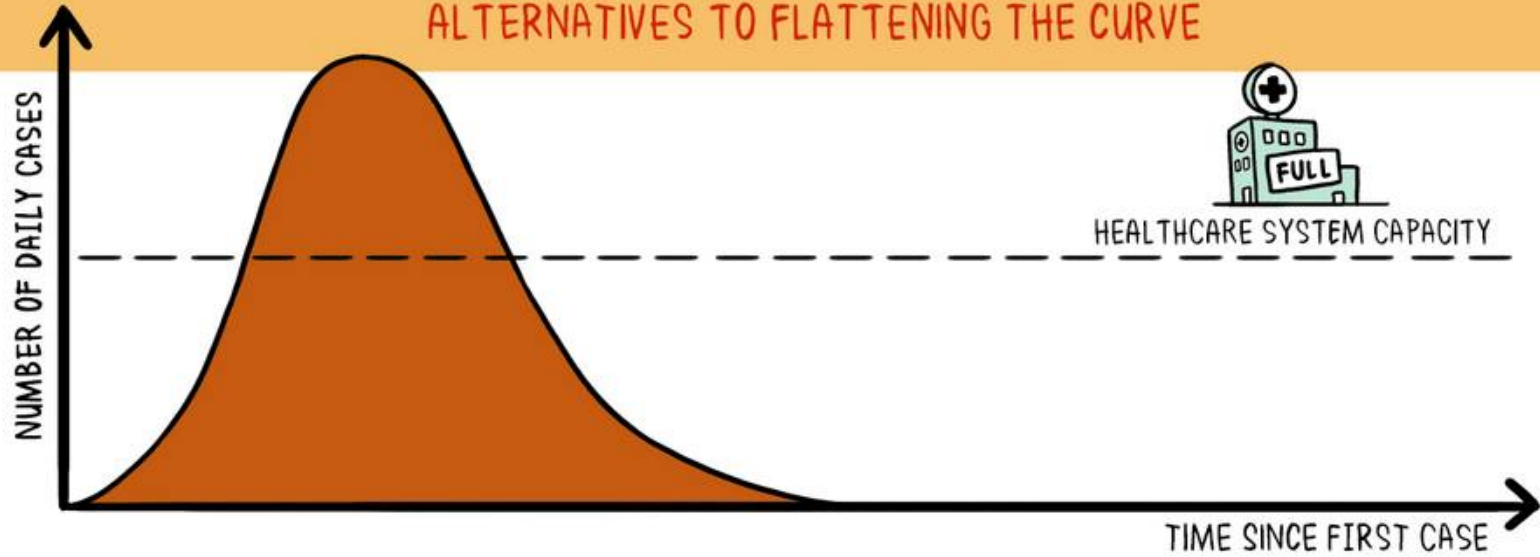
Flatten the Curve

- “flatten the surge in serious cases” to a level that will not put a strain on the hospital system.
- Governor Cuomo said [New York] state will need:
 - 110,000 hospital beds – they have 53,000, 83% of which are occupied.
 - 20,000-37,000 ICU beds – they have 3,000, 80% of which are occupied.
 - they have 10,000 ventilators
 - looking to buy more, but was beat to the punch by China, South Korea and Italy.

Cannot Give Up

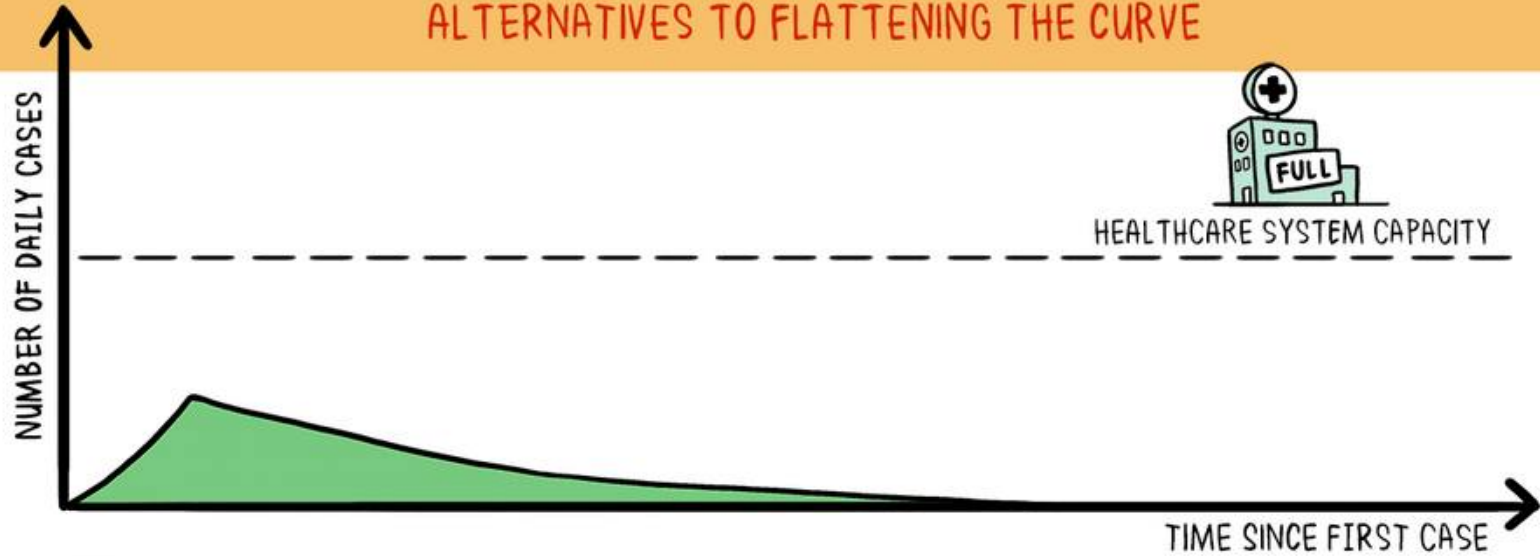
STOP THE SPREAD

ALTERNATIVES TO FLATTENING THE CURVE



STOP THE SPREAD

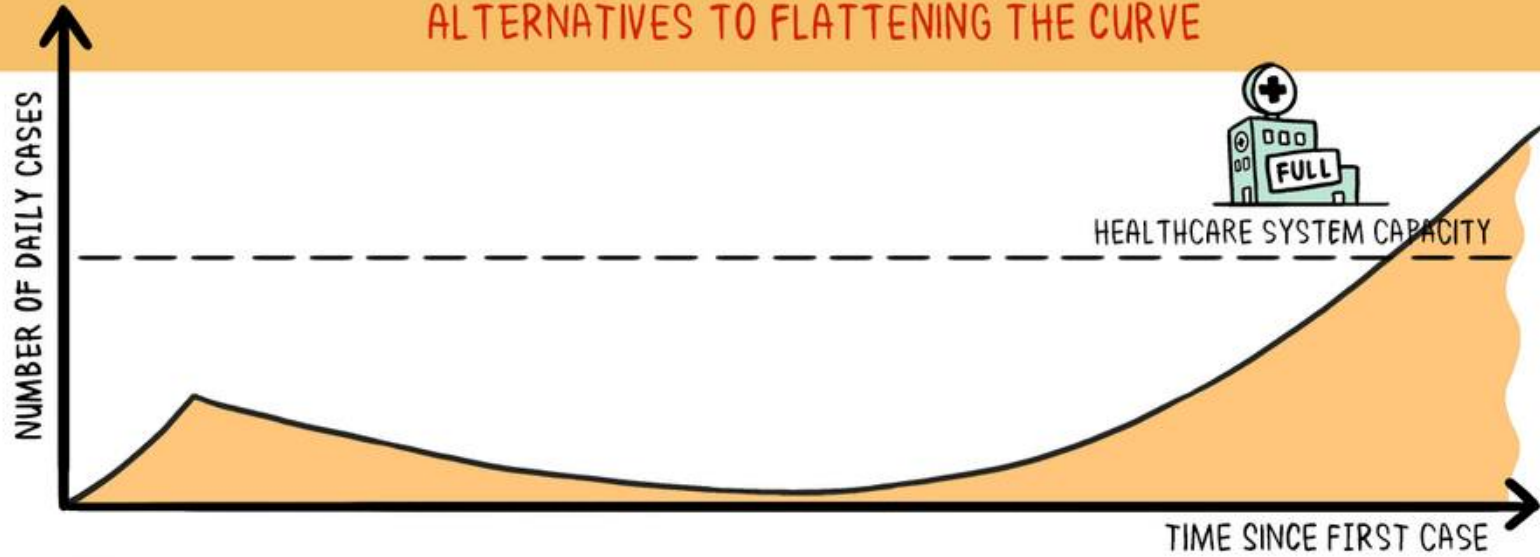
ALTERNATIVES TO FLATTENING THE CURVE



STRONG COLLECTIVE RESPONSE (BEST CASE SCENARIO)

STOP THE SPREAD

ALTERNATIVES TO FLATTENING THE CURVE



STRONG COLLECTIVE RESPONSE (SHORT TERM ONLY)

9. The CDC classifies infectious agents by four “Biosafety Levels” (BSL)

Movie Clip:
Outbreak (1995)
BSL labs

Biosafety Labs

- BSL1: agents not known to consistently cause disease in healthy adults
 - PPE: basic safety, no special equipment
- BSL2: agents that pose a danger if accidentally inhaled, swallowed, or exposed to the skin
 - PPE: gloves, eyewear, handwashing sinks, and waste decontamination facilities
- BSL3: agents that can be transmitted through the air and cause potentially lethal infection
 - PPE: gas-tight enclosure, clothing decontamination, sealed windows, and specialized ventilation systems
- BSL4: agents that pose a high risk of life-threatening disease for which no vaccine or therapy is available
 - PPE: wear full-body, air-supplied suits and shower when exiting

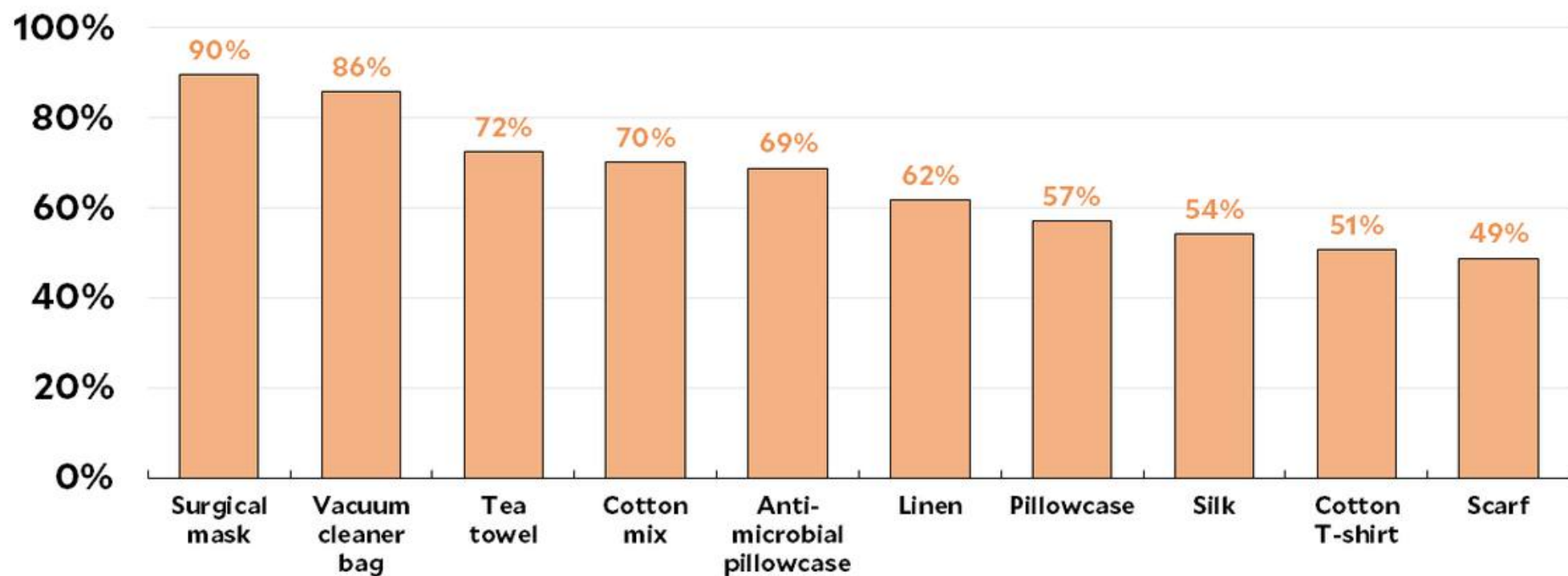
<https://www.niaid.nih.gov/research/biodefense-biosafety-labs>

Movie Clip:
Outbreak (1995)
suit tear

Movie Clip:
Contagion (2011)
BSL4 suits

Effectiveness of Homemade Masks

Filtration Efficiency Against Virus-Sized Particles



Source: "Testing the Efficacy of Homemade Masks," 2013

Mother Jones

10. Why did the CDC say they cannot grow the virus in cells? What type of cells were successful?

- CDC couldn't grow the virus in cells because they were using BSL-4 and the cells were dying.
 - Sussman was able to grow the virus in a bat cell line and was successful using BSL-3.
- The CDC cannot grow the virus in cells because it has killed every cell that it has been put in.
 - The cells that were successful in growing the virus were a **fetal bat lung** cell line.

11. When the virus mutates, Dr. Hextail says the initial R_0 for the virus was _ and now it is _

- It was 2 and now is no less than 4.
 - Without the vaccine, anticipate 1 in 12 to contract the virus.

12. The first attempt at a vaccine using dead virus was unsuccessful, what type of virus was used in subsequent attempts?

- They tried using dead virus, it did not work.
- The type of virus used in subsequent attempts was a live virus that can change back to wild type.
- a live attenuated virus

Movie Clip:
Contagion (2011)
vaccine

12. The first attempt at a vaccine using dead virus was unsuccessful, what type of virus was used in subsequent attempts?

- **Ellis:** I thought you said that once we could grow it, we could vaccinate against it.
- **Dr. Hextall:** We tried using dead virus combined with several atrophins to boost immune response.
- and?
- No protective antibodies, a lot of dead monkeys.
- Can you get to the part where there's good news?
- Now we have to try a live attenuated virus.
- Like with polio?
- Exactly. the only danger with a live virus is the possibility that it will revert to wild-type and kill the host.
- and when will we know about that?
- I'll ask the monkeys.

14. Did forsythia prove to be a treatment for individuals infected with the virus? Explain.

- No, forsythia was not an effective treatment.
- There was no scientific evidence that this ever worked.
- The [blogger] faked his symptoms and was using this as a scheme to make money.



Chloroquine

Hydroxychloroquine

- Trump said drug has "tremendous promise"
- "I think it's going to be great," he said. "We're quickly studying this drug."
- used to treat malaria, lupus, and rheumatoid arthritis



Untested Drug

- An early study from France last week suggested chloroquine could be helpful, but it was carried out on small numbers of patients and without the rigor of a randomized control trial.
- Another small trial from China, published on Tuesday, showed the drug had no impact.
- Larger trials — including one in New York and a multinational program run by the WHO — are just starting.

States Say Some Doctors Stockpile Trial Coronavirus Drugs, for Themselves

- Doctors are hoarding medications by writing prescriptions for themselves
 - states have issued emergency restrictions or guidelines on how the drugs can be dispensed at pharmacies
- Hydroxychloroquine becomes a Schedule H1 drug as hoarding leads to shortage
 - which means it cannot be sold over the counter

- <https://www.nytimes.com/2020/03/24/business/doctors-buying-coronavirus-drugs.html>



Aquarium cleaner treatment?

- A man thought aquarium cleaner with the same name as the anti-viral drug chloroquine would prevent coronavirus. It killed him.
 - The fish tank solvent that treats aquatic parasites contains the same active ingredient as the drug, but in a different form that can poison people
- The couple in their 60s reportedly poured some of the fish tank cleaning chemical, **chloroquine phosphate**, into soda and drank it.
 - They felt dizzy and started vomiting. The husband died at the hospital, and the wife is under critical care, according to the statement.
- <https://www.washingtonpost.com/nation/2020/03/24/coronavirus-chloroquine-poisoning-death/>



Other Online Hoaxes

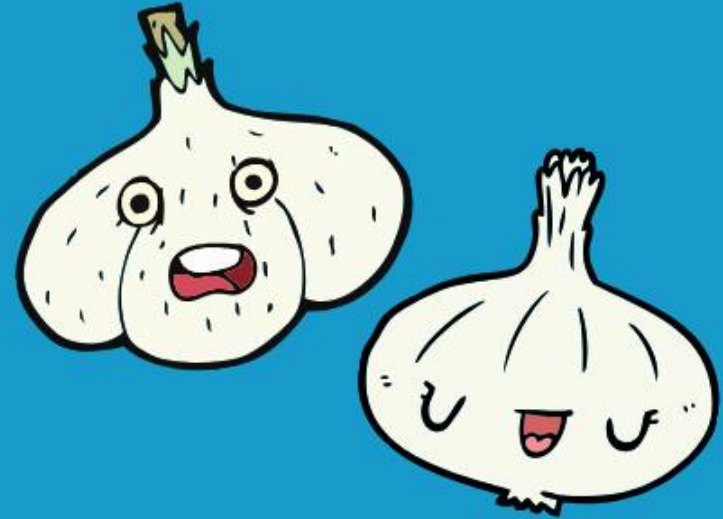
- Online hoaxes have inaccurately and dangerously suggested:
 - drinking bleach
 - dousing skin with chlorine
 - consuming colloidal silver
- They've also promoted largely harmless, but ultimately ineffective measures like:
 - gargling salt water
 - eating garlic
 - taking a hot bath
- One confused Florida official advised the public to **use hair dryers** to try to kill the virus, but later apologized.
- <https://www.who.int/emergencies/diseases/novel-coronavirus-2019/advice-for-public/myth-busters>



WHO has busted more than a dozen myths about how to prevent, treat or cure the virus.

Garlic is a healthy food that may have some antimicrobial properties. **However, there is no evidence from the current outbreak that eating garlic has protected people from the new coronavirus (2019-nCoV)**

Can eating garlic help prevent infection with the new coronavirus?



Censured Indian Plant Gets USFDA Nod to Supply Trump-Touted Drug

- lifted import restrictions in an effort to increase supplies of a malaria drug
- Three manufacturing plants get exemption
- Agency had cited ‘systematic data manipulation’ at one plant
 - inspectors discovered multiple violations of its manufacturing guidelines, including “systemic data manipulation” in tests meant to ensure the drugs’ efficacy and safety
- <https://www.bloomberg.com/news/articles/2020-03-23/fda-lifts-import-curbs-on-maker-of-unproven-virus-drug-in-india>

Global rush for coronavirus drug sees India stop exports

- India has **stopped exporting** the malaria drug hydroxychloroquine
 - despite the FDA lifting its ban on imports due to its potential as a coronavirus treatment
- India is prioritizing the supply of a potential coronavirus drug to its own people
- <https://www.ft.com/content/dc8386ba-6eae-11ea-89df-41bea055720b>



Summary 2. What are the symptoms of individuals infected with the virus?

- Cold and Cough, High fever, also Seizures (epilepsy)

Summary 4. How realistic do you think this film was?

- **Meghan:** I think that the way the CDC and WHO worked to find a vaccine was accurate. However, a vaccine would never be able to be available to the public that fast without having to do multiple clinical trials first.
- Also, as we can see today nobody is looting stores, at least not yet.
- The way that the disease spread so fast and how people were slow to react I believe is realistic.
- I think that basing who gets the vaccine first based on birth dates would not go over well in this society. People would lie or give their vaccine up to a family member.

Summary 4. How realistic do you think this film was?

- **Mohammad:** I felt it was fairly realistic overall. The incidence and spread seems to be the same as it is in today's situation with COVID-19.
- But the making of a vaccine and how it's going to be distributed was far from realistic. No doctor is going to inject their own self as a test subject or be able to go through the process without regulations and test groups.
- The representation of how society would react was perfect.

Sources

Online Sources for Contagion

- “The Smartest (and Dumbest) Movies to Watch During an Outbreak” by Angela Watercutter at WIRED.com
 - <https://www.wired.com/story/pandemic-movies-tv/>
- “For Me, Rewatching ‘Contagion’ Was Fun, Until It Wasn’t” by Wesley Morris at NY Times
 - <https://www.nytimes.com/2020/03/10/movies/contagion-movie-coronavirus.html>
- “Fact-Checking ‘Contagion’ — In Wake Of Coronavirus, The 2011 Movie Is Trending” by Goats and Soda at NPR
 - <https://www.npr.org/sections/goatsandsoda/2020/02/16/802704825/>
- “How the makers of ‘Contagion’ saw an outbreak like coronavirus coming” by Soumya Karlamangla at LA Times
 - <https://www.latimes.com/california/story/2020-03-11/coronavirus-contagion-outbreak-accuracy-movie>
- “A Scientific Adviser On “Contagion” Said People Should Have Taken The Film More Seriously” by Krystie Lee Yandoli at BuzzFeed News
 - <https://www.buzzfeednews.com/article/krystieyandoli/contagion-movie-coronavirus>
- The Ending of Steven Soderbergh’s Contagion, Revisited
 - <https://www.vulture.com/2020/01/contagion-movie-ending-coronavirus-and-pandemic-panic.html>

Online Sources

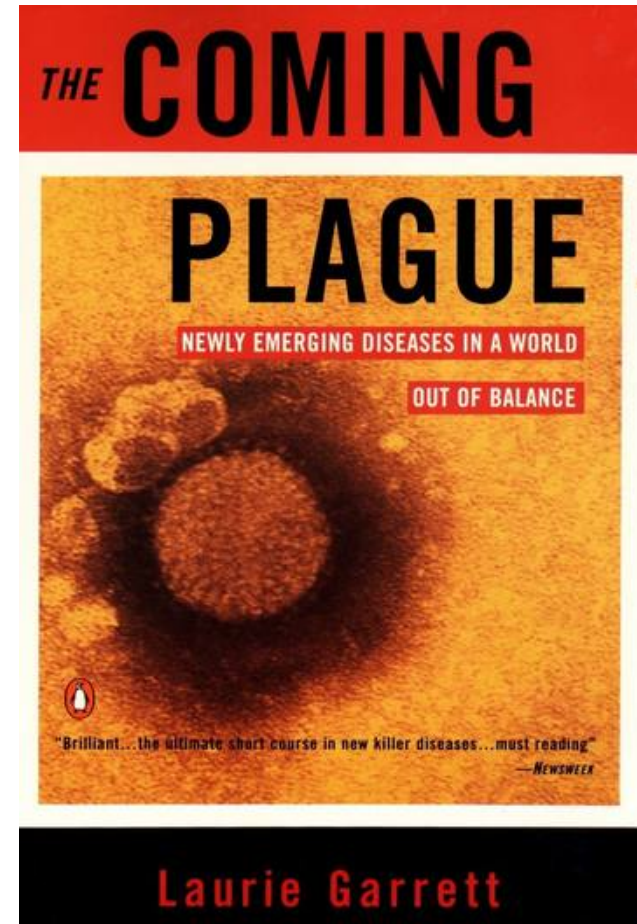
- 13 Coronavirus myths busted by science
 - <https://www.livescience.com/coronavirus-myths.html>
- R0: How Scientists Quantify the Intensity of an Outbreak Like Coronavirus and Its Pandemic Potential
 - <https://sph.umich.edu/pursuit/2020posts/how-scientists-quantify-outbreaks.html>
- Report of the WHO-China Joint Mission on Coronavirus Disease 2019 (COVID-19)
 - <https://www.who.int/docs/default-source/coronaviruse/who-china-joint-mission-on-covid-19-final-report.pdf>
- CDC Information Page:
 - <https://www.cdc.gov/coronavirus/2019-nCoV/index.html>

Science Publications

- Clinical course and risk factors for mortality of adult inpatients with COVID-19 in Wuhan, China: a retrospective cohort study
 - [https://doi.org/10.1016/S0140-6736\(20\)30566-3](https://doi.org/10.1016/S0140-6736(20)30566-3)
- The Incubation Period of Coronavirus Disease 2019 (COVID-19) From Publicly Reported Confirmed Cases: Estimation and Application
 - <http://dx.doi.org/10.7326/M20-0504>
- Severe acute respiratory syndrome-related coronavirus: The species and its viruses – a statement of the Coronavirus Study Group
 - <https://doi.org/10.1101/2020.02.07.937862>

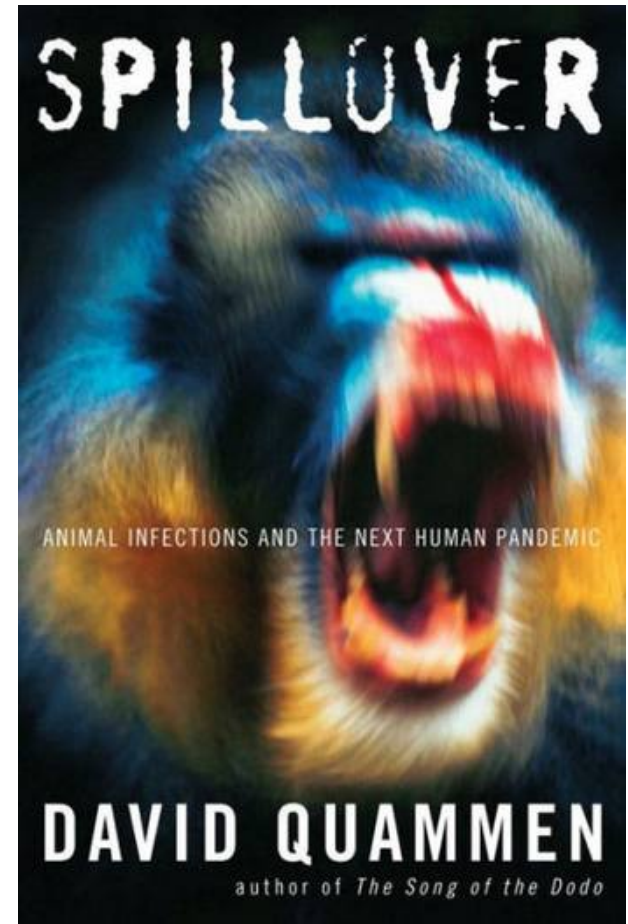
Literature

- “The Coming Plague: Newly Emerging Diseases in a World Out of Balance” by Laurie Garrett



Literature

- “Spillover: Animal Infections and the Next Human Pandemic” by David Quammen



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